LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	32
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of inertial navigation system and integration with MHRS system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of inertial navigation system and integration with MHRS system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS integration diagrams; Instructor supervision required for system logic verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an Inertial Navigation System?
(b) How does the AFCS utilize inertial data for aircraft control?
(c) What is the role of the MHRS in integrating navigation systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic principles of inertial navigation • Components of the inertial platform • Error accumulation in INS • INS integration with MHRS architecture • Data flow between INS and AFCS	L & Ds	PPT, IB	15 min
(b)	• AFCS control loop mechanisms • Servo actuator response to inertial inputs • MHRS fault detection and isolation • Cross-checking INS data with other sensors • Impact of INS errors on flight path	L & Ds	PPT, PDF	20 min
(c)	• System redundancy in MHRS • Maintenance procedures for INS units • Calibration requirements for inertial sensors • Troubleshooting common INS integration faults • Operational limitations of INS in specific environments	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the AFCS uses inertial data to maintain aircraft attitude.
(b) Describe the process by which the MHRS detects an INS failure.
(c) What are the primary sources of error in an inertial navigation system?

12. Home Assignment and Follow Up	Prepare a brief report on the integration of the inertial navigation system with the MHRS, focusing on data exchange protocols and redundancy measures, to be submitted in the next lesson (LP 02).

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and explanation of electro pneumatic servo actuators and electro mechanical servo actuators.	Refer to the master syllabus reference list	LP 33

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min